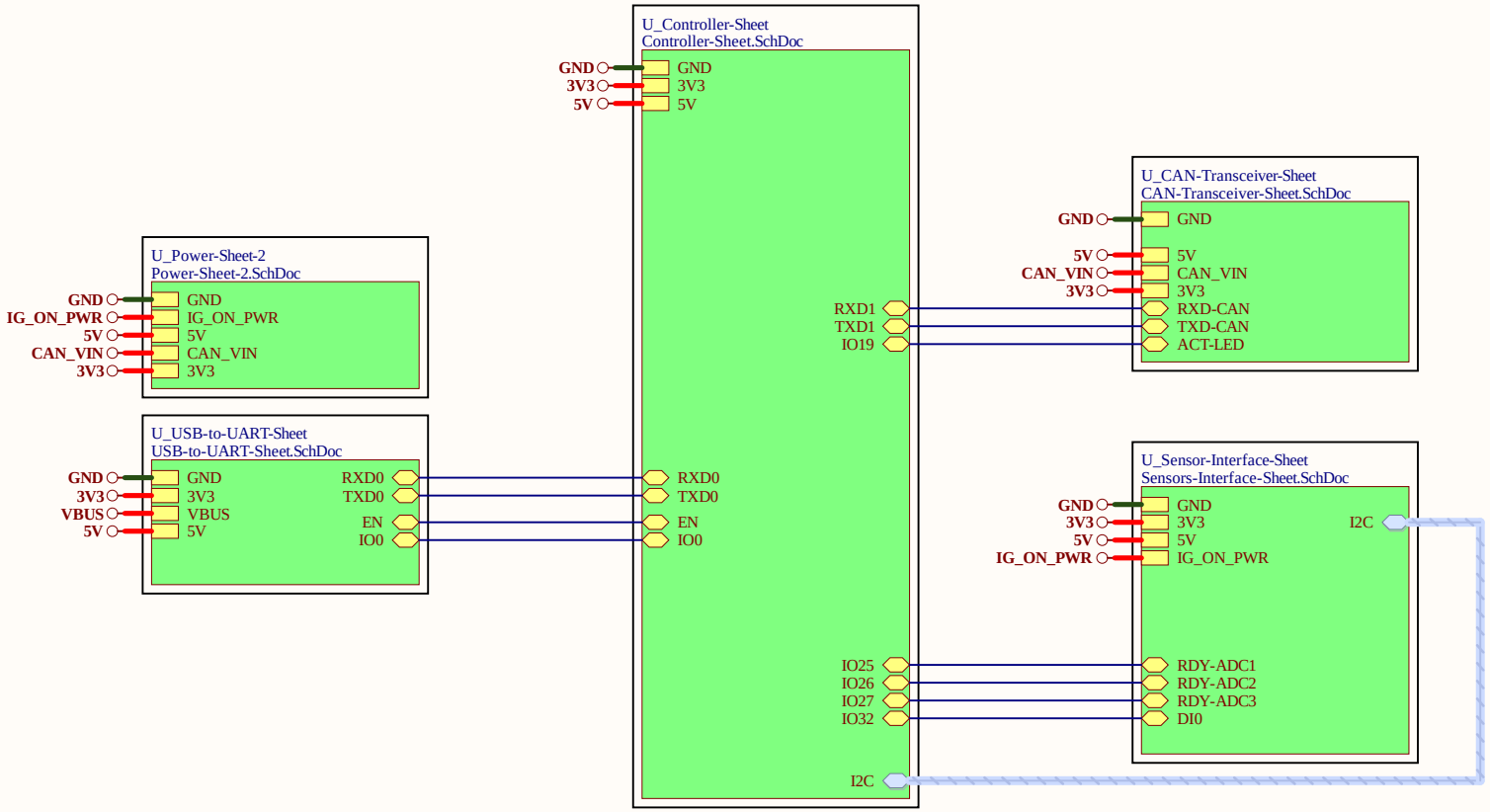
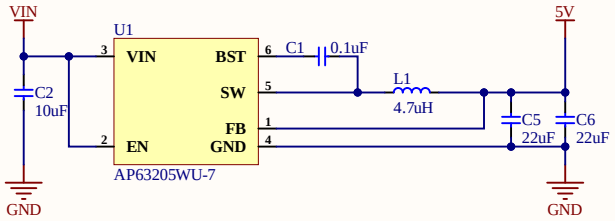
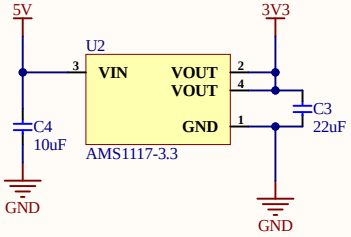


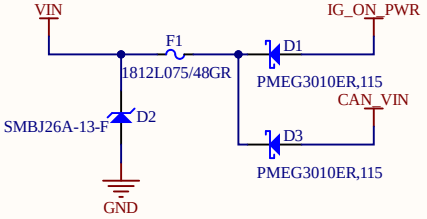
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Date: 2/10/2025	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 6

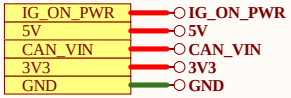








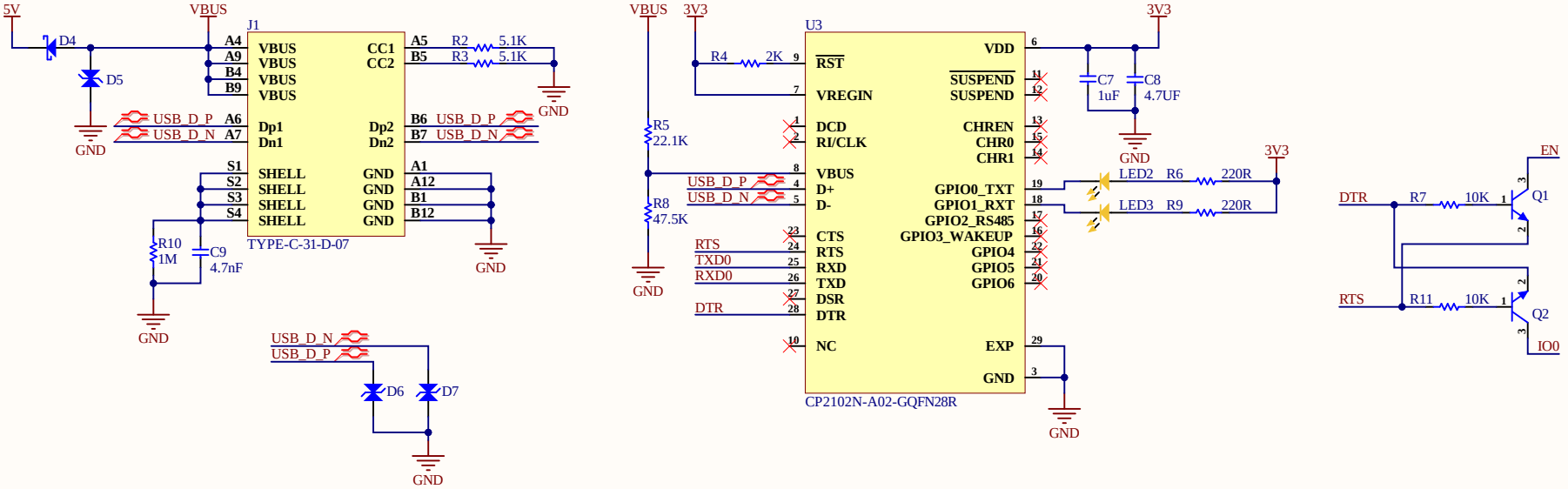
VIN (9VDC-30VDC) TO 5V & 3V3 BUCK CONVERTER



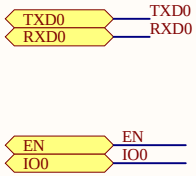
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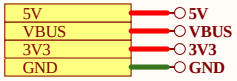
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Date: 2/10/2025	Page	Num: 2
Drawn By: Huzaifa Ali	Page	Total: 6

USB CONNECTOR FOR DEBUGGING



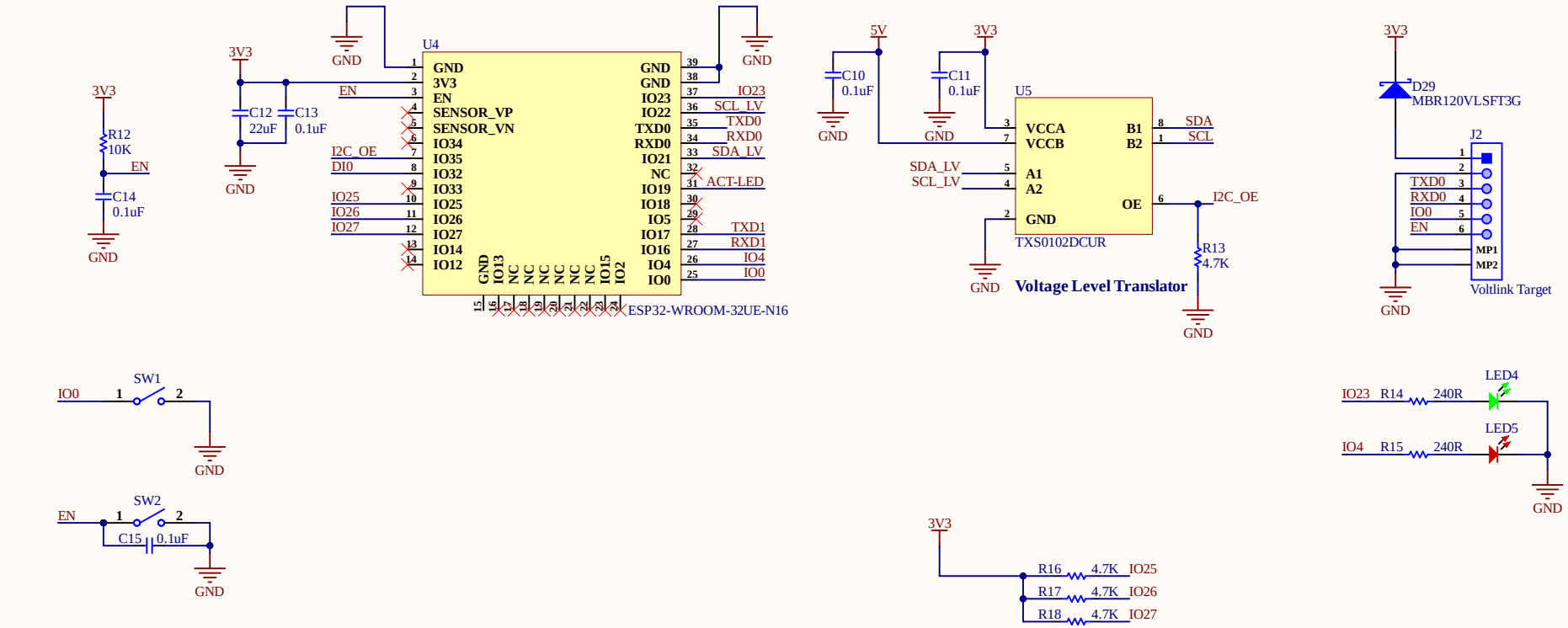
USB-TO-UART



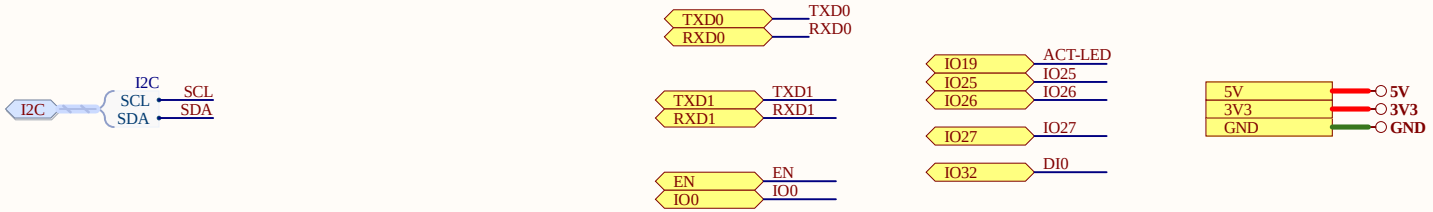


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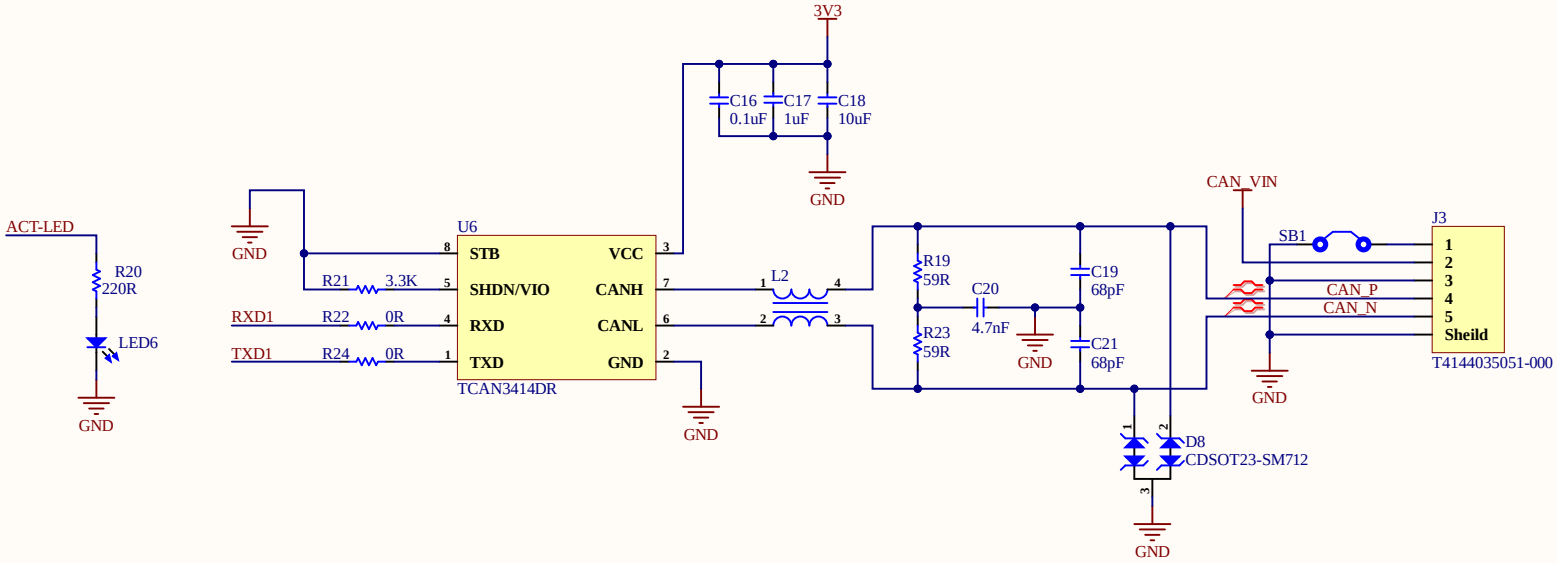


CONTROLLER SHEET



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Title: Controller-Sheet.SchDoc	
File: NMEA2K-Converter-v1.0.PrjPcb	
Date: 2/10/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 6



CAN TRANSCIEVER NMEA2000

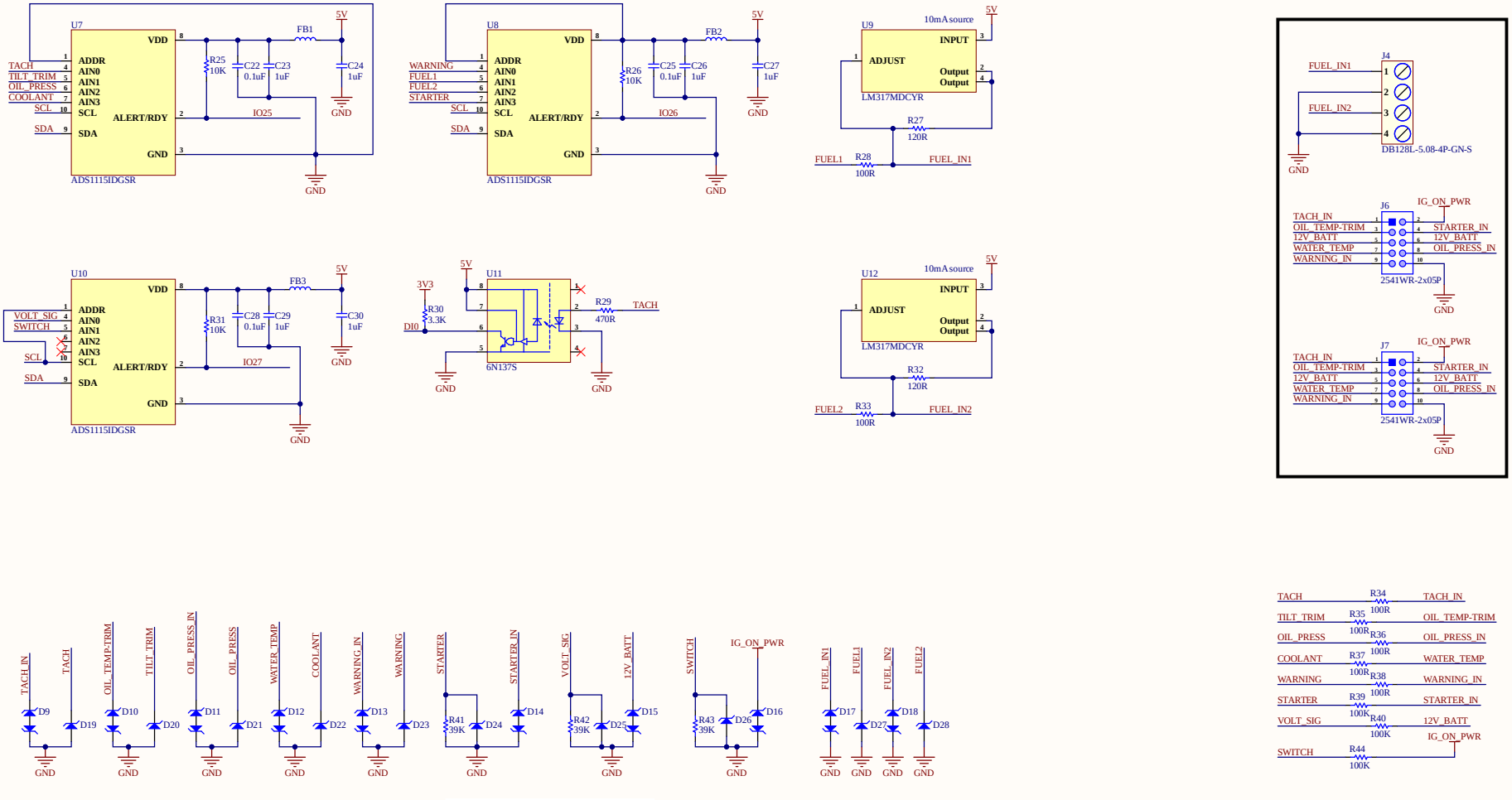
RXD-CAN — RXD1
TXD-CAN — TXD1

ACT-LED — ACT-LED

5V — 5V
CAN_VIN — CAN_VIN
3V3 — 3V3
GND — GND

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: CAN-Transceiver-Sheet.SchDoc	
File: NMEA2K-Converter-v1.0.PrjPcb	
Date: 2/10/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 6



SENSOR INTERFACE SHEET

DI0

RDY-ADC1
RDY-ADC2
RDY-ADC3
IO25
IO26
IO27

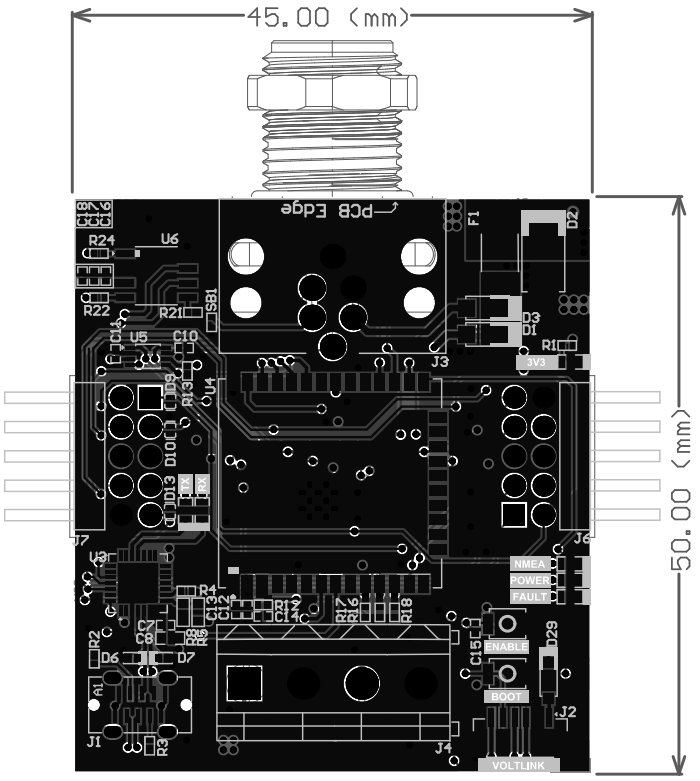
I2C
SCL
SDA

USE FSR = 6.144V
ADDR_U8 -> GND
ADDR_U9 -> VDD
ADDR_U10 -> SCL

IG_ON_PWR
5V
3V3
GND

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Sensors-Interface-Sheet.SchDoc	
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Date: 2/11/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 6



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.40mil	3.5	
1	Top Layer		1.38mil		
	Dielectric 2	PP-006	8.28mil	4.1	
2	Layer 1	CF-004	0.60mil		
	Dielectric 1	FR-4	41.93mil	4.8	
3	Layer 2	CF-004	0.60mil		
	Dielectric 3	PP-006	8.28mil	4.1	
4	Bottom Layer		1.38mil		
	Bottom Solder	Solder Resist	0.40mil	3.5	
	Bottom Overlay				
Total board thickness:			63.25mil		